Himanshu Tiwari | himu.me

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EXPERIENCE

Project Coordinator, Network Delivery Jul 2026 – Present
StarTech Networks Inc *Remote · US Operations*

- Coordinate telecom modernization workflows involving Ericsson BBU hardware, asset lifecycle management, site migrations, and field operations across US markets for Verizon Wireless.
- Consolidate and validate large-scale operational records, helping eliminate a year-long backlog of pending network data.
- Develop scripting and AI-assisted workflows to automate repetitive operational tasks.

Research Fellow Jun 2025 – Jul 2025
LIA Lab, CERI — Université d'Avignon *Avignon, France*

- Engineered LoRaPwn, a custom LoRaWAN security framework with firmware-level MAC implementation and direct LoRa PHY control.
- Developed STM32 HALs in Rust and bare-metal drivers, bypassing vendor abstraction stacks.
- Replaced AES-128 with ChaCha20-Poly1305 AEAD, achieving 10× faster execution and 90% lower cryptographic energy consumption.
- Ranked top fellow in the 2025 cohort with a 20/20 jury evaluation.

Head of Embedded Systems Oct 2024 – May 2025
Pravideon Pvt. Ltd *Navi Mumbai, India*

- Developed Nirvanagni, an autonomous fire suppression system using TensorFlow-based detection and Modbus PLC integration; trained on 150K images.
- Automated P&ID diagram analysis using computer vision and AI, winning 1st place at Echelon'25.

PROJECTS

Ginti | Go · Linux | github.com/HimuCodes/ginti

- Engineered a high-performance `wc` replacement processing 1.1GB in 0.25s (~10× GNU wc) using concurrency, SIMD, `mmap`, and cache-aware I/O.

Slipper Zero | C · ESP-IDF · Wireshark | https://doi.org/10.1109/gcwot63882.2024.10805625

- Engineered an ESP32 wireless security platform for packet capture, Wi-Fi analysis, and PCAP processing.
- Published the research in IEEE Xplore and presented it at GCWOT 2024, Spain.

TECHNICAL SKILLS

Languages: Go, Python, Embedded C, Bash
Systems: Linux, Git, Docker, Networking, Concurrency, Performance Optimization
AI: TensorFlow, Computer Vision, LLM APIs, AI Agents, Harness
Embedded: STM32, ESP32, PCB Design, UART, SPI, I2C, LoRaWAN, Modbus

EDUCATION

University of Mumbai
B.E. Electronics & Telecommunications, CGPI: 7.47/10 *Nov 2022 – Jun 2026*

ACHIEVEMENT

IEEE Publication: Slipper Zero IEEE Xplore, GCWOT 2024 | https://doi.org/10.1109/gcwot63882.2024.10805625
Hackathons: 3× ViSION Project Exhibition Winner · Echelon'25 Winner
Buildspace: Selected from **32,000+ global applicants** for Nights & Weekends S5.
Leadership: Industry Relations Manager, Training & Placement Office